THE UNIVERSITY OF QUEENSLAND
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Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Mathematics & Physics
Semester Two Examinations, 2023
PHYS7045 Quantum Technologies

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•This is an Open Book examination
•Any calculator permitted - unrestricted
•During Planning Time - Students are encouraged to review and plan responses to the exam questions
•This examination paper will be released to the Library

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

* Open-book: Any additional written or printed material is permitted; material may also be annotated.

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

2 x 14-Page Answer Booklet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

For Examiner Use Only

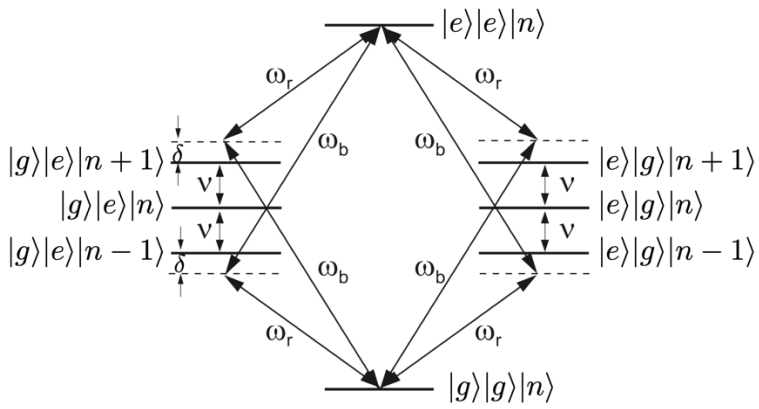
Question	Mark

Total _____

Question 1.

The Mølmer-Sørensen gate is often used in ion trap quantum computing as its operation is independent of the vibrational state of the ion, so does not require cooling to the motional ground state.

The scheme is shown below and consists of a bichromatic (two frequencies) laser interacting with each of the ions. This results in a two-photon transition from the ground $|g\rangle|g\rangle|n\rangle$ state, where $|n\rangle$ represents the phonon occupation, to the doubly excited state $|e\rangle|e\rangle|n\rangle$. The frequencies ω_r and ω_b drive the 1st red and blue sidebands respectively, and the lasers are detuned by $\pm\delta$ from the transitions so that the intermediate states are not extensively populated. The frequency ν is the vibrational transition frequency.



Part A (grade 1-5)

1. Focusing on a single ion, and for the transitions from the ground state to the intermediate states only, show the effect of the red and blue sideband transitions and determine their respective Rabi frequencies.

2. Completing the transitions to the $|e\rangle|e\rangle|n\rangle$ state and noting that a two-photon Rabi frequency is $(\Omega_1\Omega_2)/2\delta$, determine the two photon Rabi frequency for the upper and lower two photon transitions.

3. Using the formula $\Omega_{eff} = |\Omega_{upper} + \Omega_{lower}|$ show that the resulting effective Rabi frequency is independent of the vibrational state $|n\rangle$.

Part B (Advanced: grade +0, +1, +2)

(a) Determine a pulse that results in a maximally entangled Bell State of the two ions, explaining your reasoning.

Question 2.

A code defined on six qubits has the following stabilisers:

Qubit	1	2	3	4	5	6
S_1	Z	$\mathbb{1}$	Z	$\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$
S_2	X	Z	Y	Y	X	$\mathbb{1}$
S_3	X	X	X	X	Z	$\mathbb{1}$
S_4	$\mathbb{1}$	Z	Z	X	$\mathbb{1}$	X
S_5	X	Y	X	Y	$\mathbb{1}$	Z

Part A (grade: 1-5)

1. Do you expect S_1 to commute with all the other stabilisers listed in the table, or not (give reasons)? Confirm whether this is true (and show your reasoning).

2. A logical Z_L operator is defined for this code as $Z_L = X\,Y\,Y\,X\,\mathbb{1}\,\mathbb{1}$. Verify whether this commutes with the stabilisers.

3. Is the 6-qubit code able to detect a single qubit bit-flip error (i.e. an X error) on any one qubit (give reasons based on possible syndromes)? Can it correct such an error?

4. Is the 6-qubit code able to detect a single qubit bit-and-phase error (i.e. a Y error) on any one qubit (give reasons based on possible syndromes)? Can it correct such an error?

5. Explain whether the 6-qubit code can correct any single qubit X, Y or Z error. If so, what is the lower bound on the code distance?

Part B (Advanced: grade +0,+1,+2)

a) A logical X_L operator can be found of the form $X_L = A\,B\,A\,B\,\mathbb{1}\,\mathbb{1}$, where A and B are Pauli operators. Fill in the blanks to find X_L, and verify that it anti-commutes with Z_L and commutes with the stabilisers.

b) Recall that the *weight* of an operator is the number of non-identity operators in the tensor product. The weight of the logical operators presented here is 4, so the code distance satisfies $d \leq 4$. Can you find a product of stabilisers and logical operators with weight less than 4? Given part A5 above, what is the code distance?

Question 3.

A two-qubit system has the Hamiltonian:

$$H = \hbar \frac{g}{2} (Y_1 + Y_2)^2,$$

where Y_i is the Pauli y-matrix.

Part A (grade 1-5)

1. Show that the dynamics is determined by the Unitary operator

$$U(t) = \cos(gt)\, \mathbb{I} \otimes \mathbb{I} - i \sin(gt) Y_1 \otimes Y_2.$$

2. If the system starts in the initial state $|\psi(0)\rangle = |0\rangle \otimes |0\rangle$, find a time such that the state evolves to a maximally entangled state.

Part B (Advanced: grade +0,+1,+2)

a) Find a circuit that uses this entangling unitary to implement a CNOT gate.

END OF EXAMINATION